

Visualizing Biomolecules

Houston Area Biophysical Simulation
Workshop

June 15, 2023

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Before we start

- Let's download a bunch of files that we need
- Go to:

- <https://smog-server.org/workshops>

- Follow the link for “CTBP_2023”

You will see file called “vizfiles.zip”

Download this file and unzip it on your machine

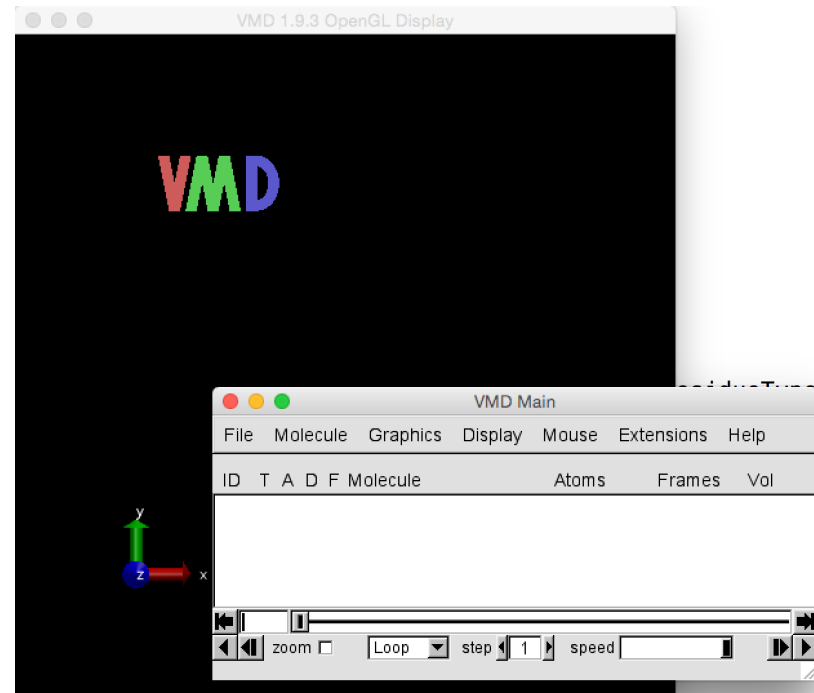
This will create a directory called “vizfiles”

Available software

- PyMol – most common with biochemists
- Chimera – tends to be preferred by cryo-EM researchers
- VMD – the only option for simulations

VMD

- “Visual Molecular Dynamics”
- Excellent package for viewing molecules and simulations of biomolecules
- Allows for many types of representations
- Should already be installed on your desktop.
- Please open it now.

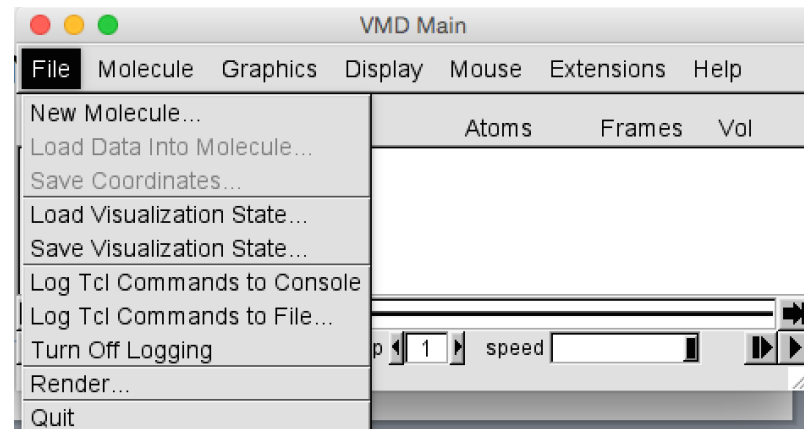


Exercise 1: Visualize a Protein

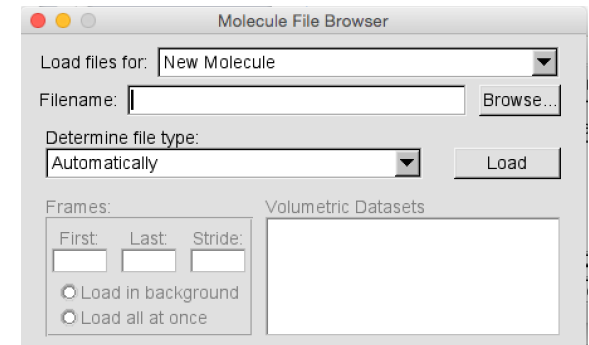
- For our first example, we will look at the structure of a protein: Adenylate Kinase (Adk).
- Adk is responsible for maintaining energy balance (ATP/ADP ratios) in the cell
- Adk has ~2000 atoms (not including Hydrogen)
- We would like to inspect the structure of this protein.

Load a Molecule

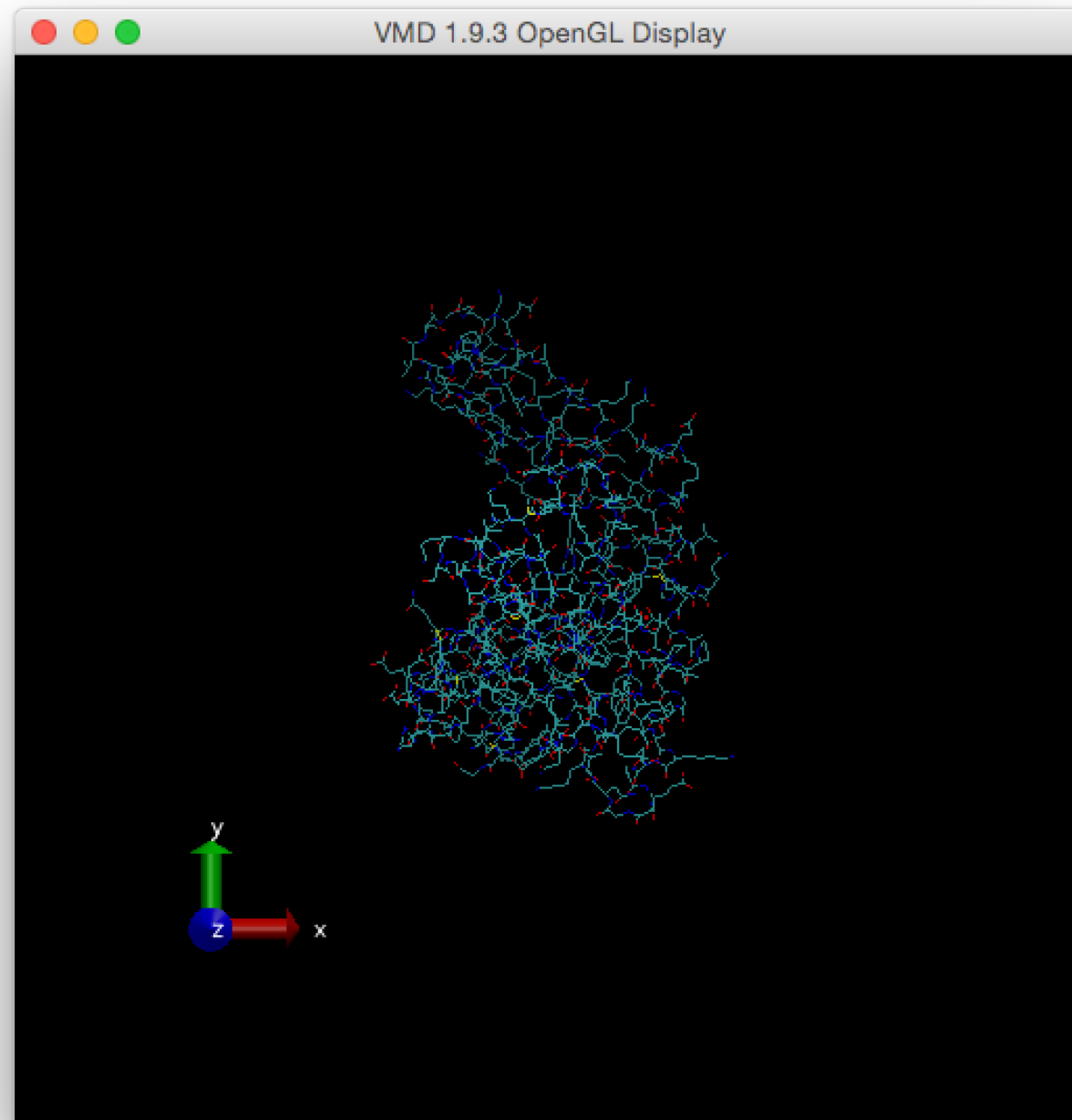
- After opening VMD, load the protein structure.
- File->New Molecule



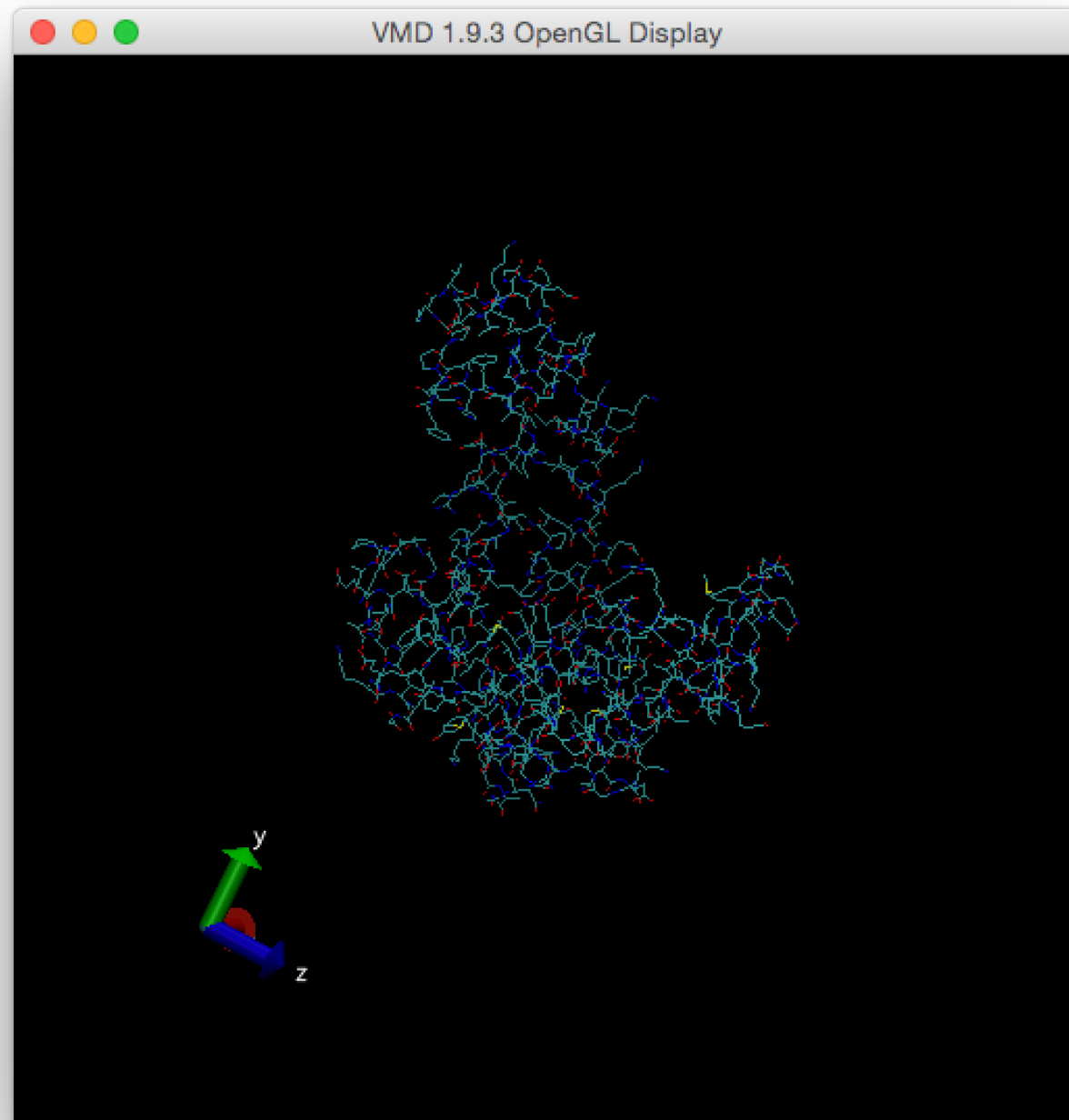
- When Molecule File Browser opens, select “Browse”.
- Navigate to the “vizfiles” directory and select the file 4AKE.pdb
- Click “Load”



You will see Adk. Each line represents a bond and each atom is shown as a vertex



Use the mouse to rotate the view of the molecule

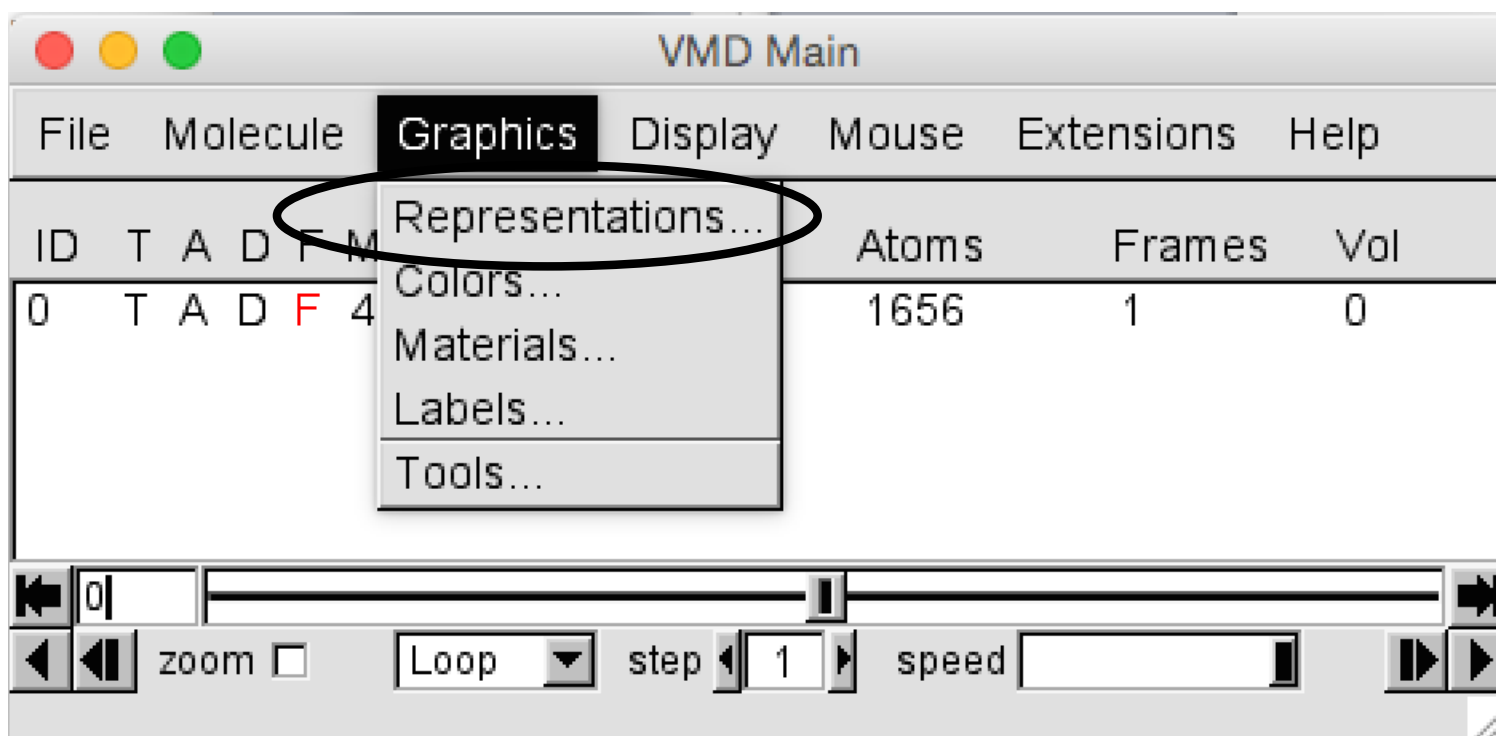


Changing the representation

- The line representation of a biomolecule is not always easy to look at.
- How can you get a feel for what the structure represents?
 - VMD provides many options

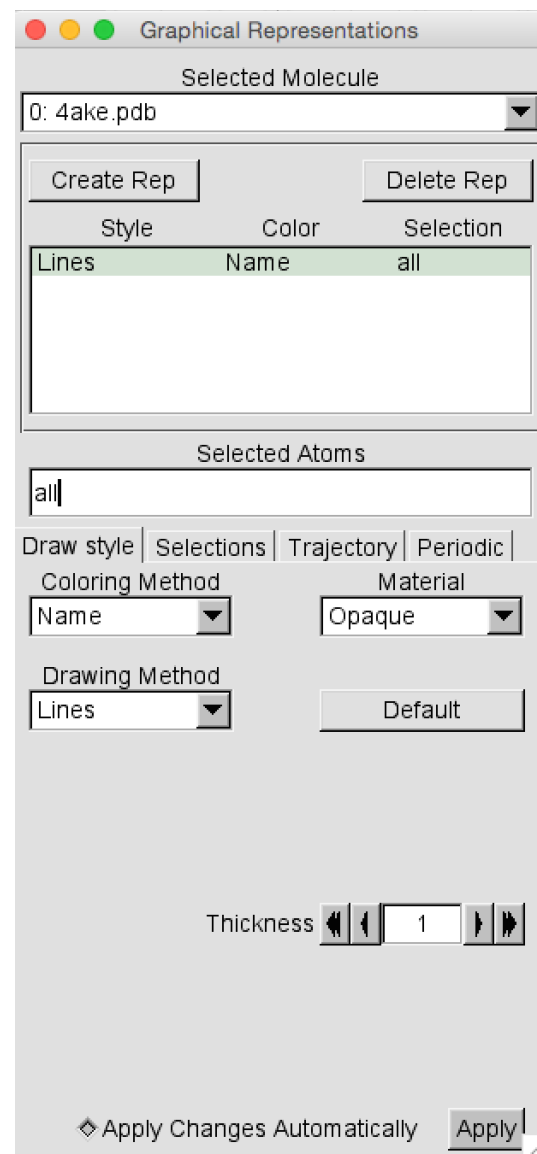
Changing the representation

- Select Graphics->Representations



Changing the representation

- Within the Graphical Representation window, use the “Coloring Method” and “Drawing Method” pull-down menus to change how the molecule is represented.
- Try some different representations



Changing the representation

Select the Drawing Method “New Cartoon”

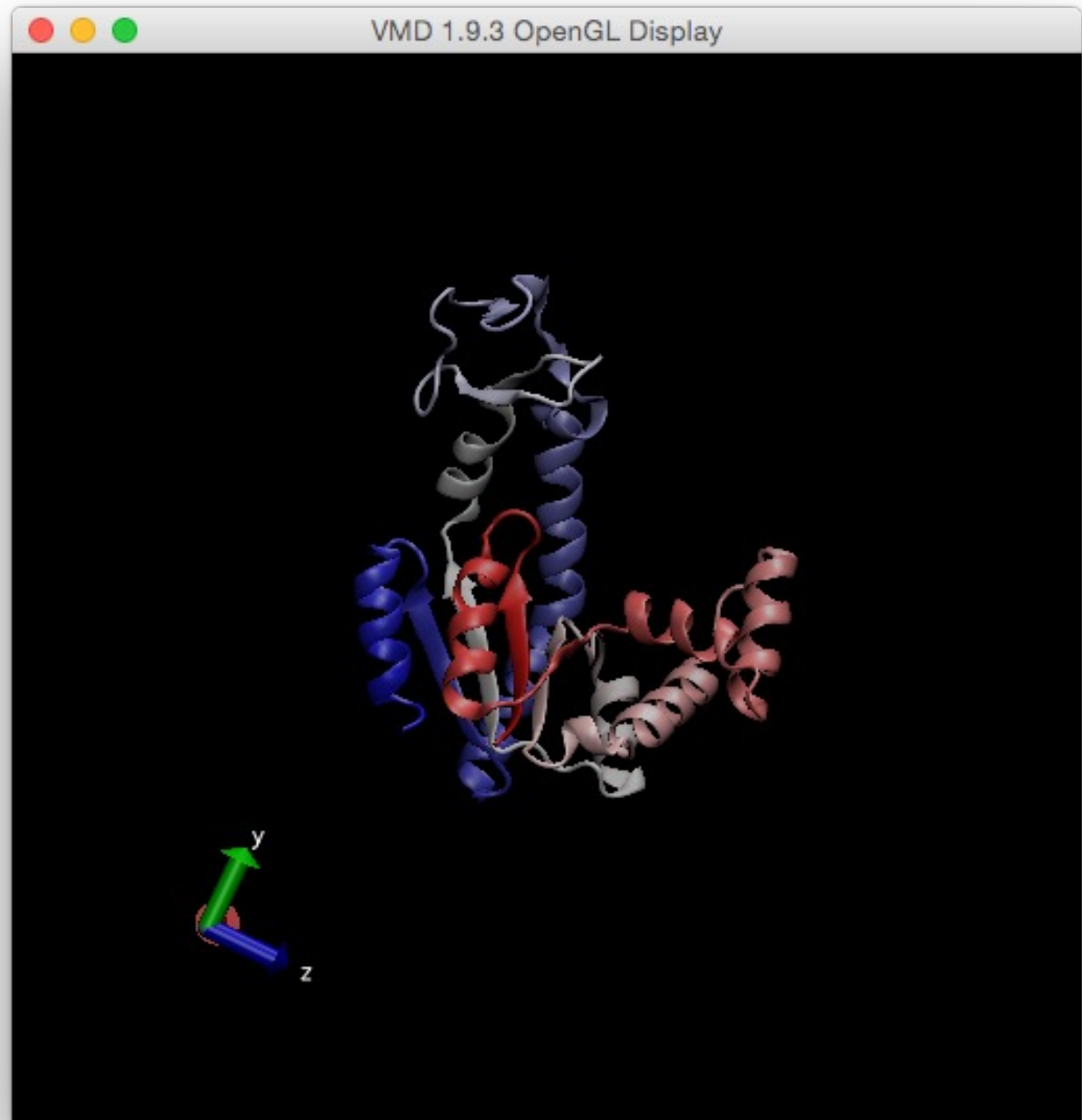
Select Coloring Method “Index”

Hit “Apply”

Different types of local structure (called secondary structure) are now visible.

Curling regions call “Alpha Helices”

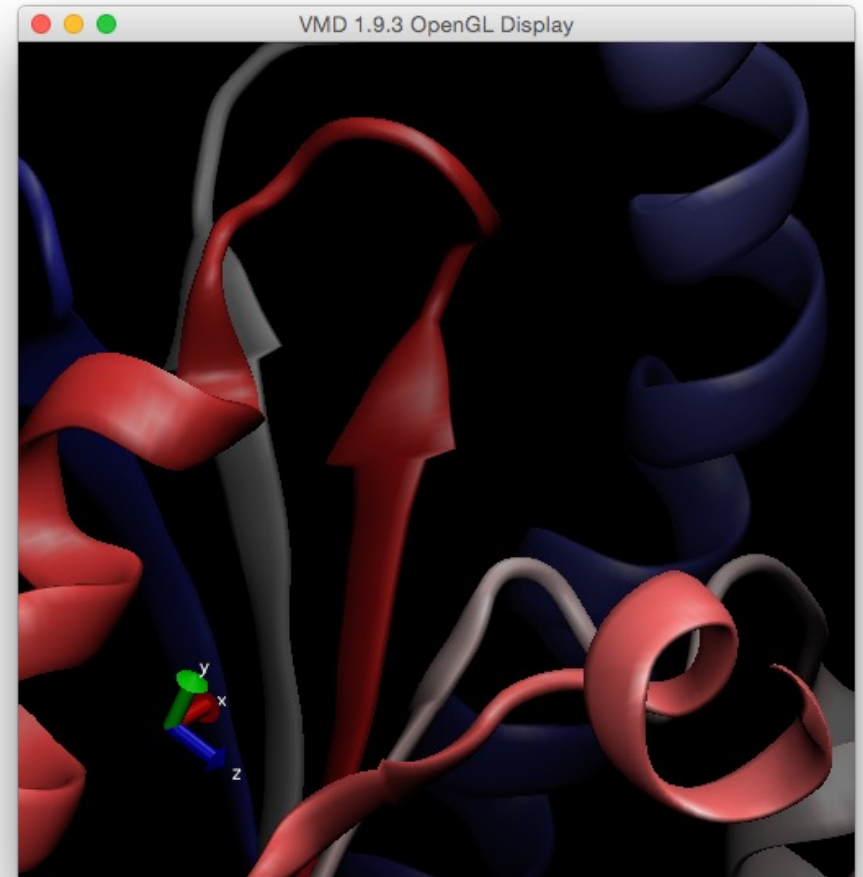
Flat regions are called “Beta Strands”



Changing the View

Try zooming in on the structure

- Select Display window
- Type “s”
 - Now you are in “scale” mode
- Click and drag the mouse to zoom in and out



Changing the Representation

Go back to the “Graphical Representation” window

Select the Drawing Method “CPK”

Select Coloring Method “Type”

Hit “Apply”

Overall structure is less obvious, but every atom may be inspected.

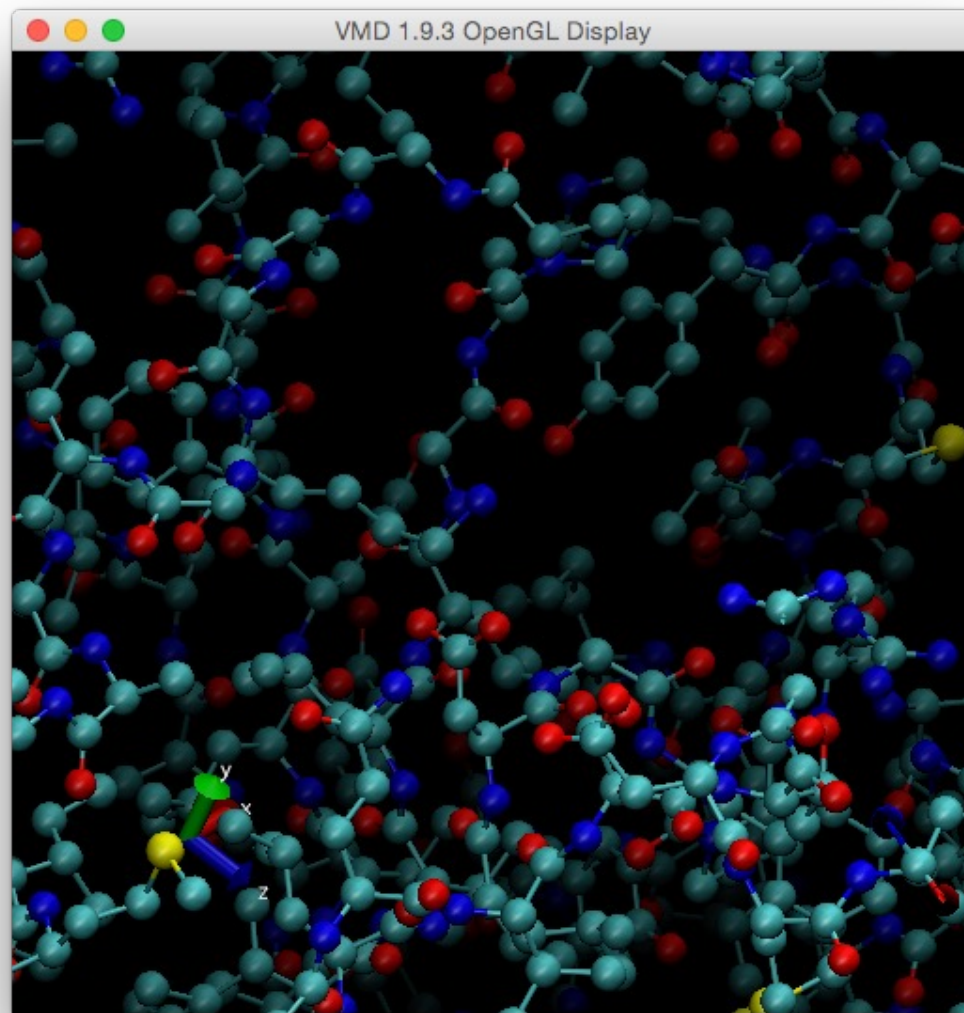
Now:

Nitrogen atoms are blue

Oxygen atoms are red

Carbon atoms are cyan

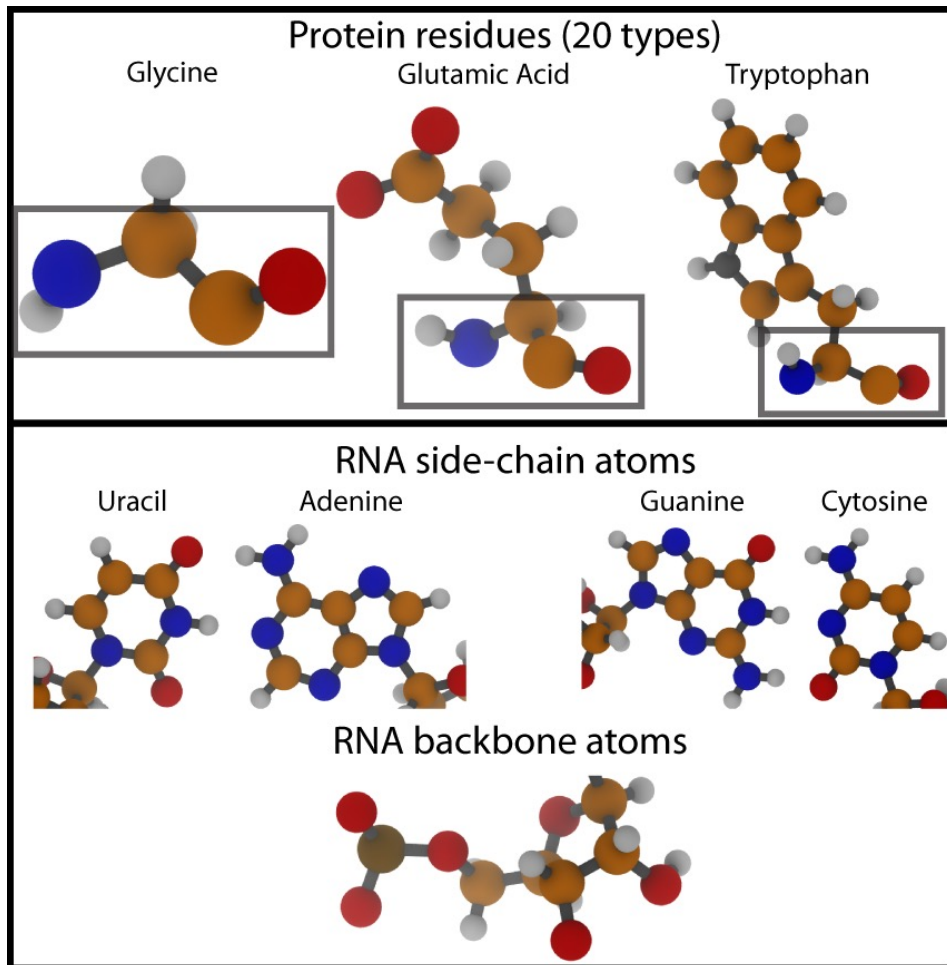
Sulfur atoms are yellow



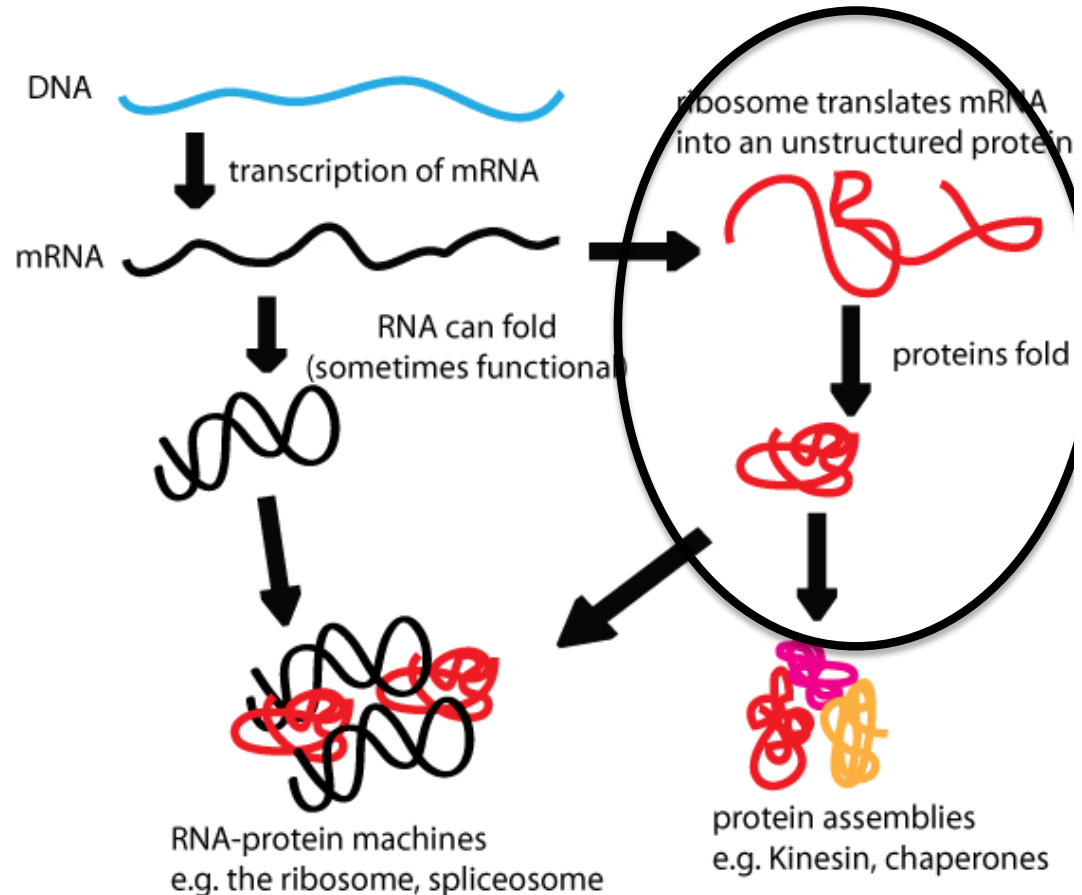
Exercise 2: Watch a Protein Fold

Making and Organizing Biomolecules in the Cell

Residues form biomolecular chains



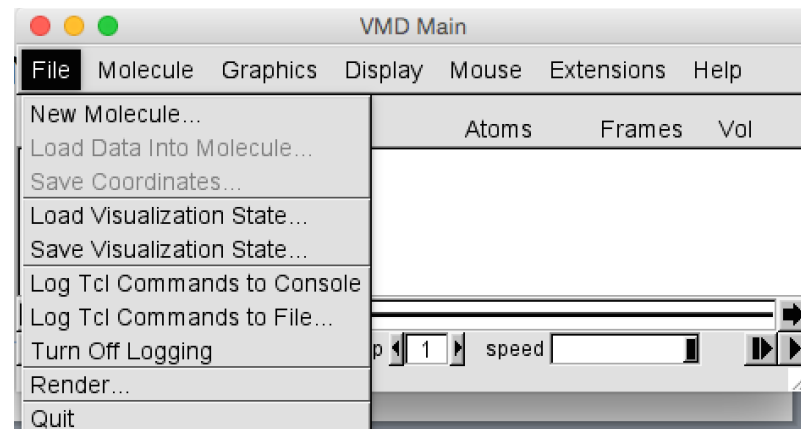
Biomolecules form functional assemblies



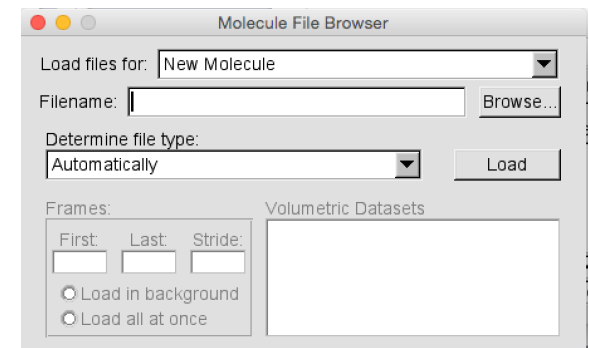
Functional aspects of molecules often require folding and assembly.
Biomolecular machines are extremely dynamic and flexible.

Watching a Simulation of Protein Folding

- After opening VMD, load the protein structure.
- File->New Molecule



- When Molecule File Browser opens, select “Browse”. In the “vizfiles” folder, select the PDB file Cl2.pdb
- Select “Load”



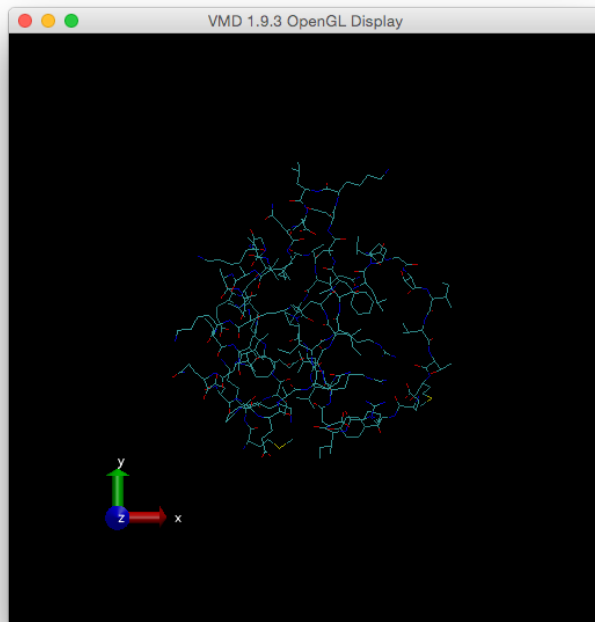
Change the Representation

Select the Drawing Method “New Cartoon”

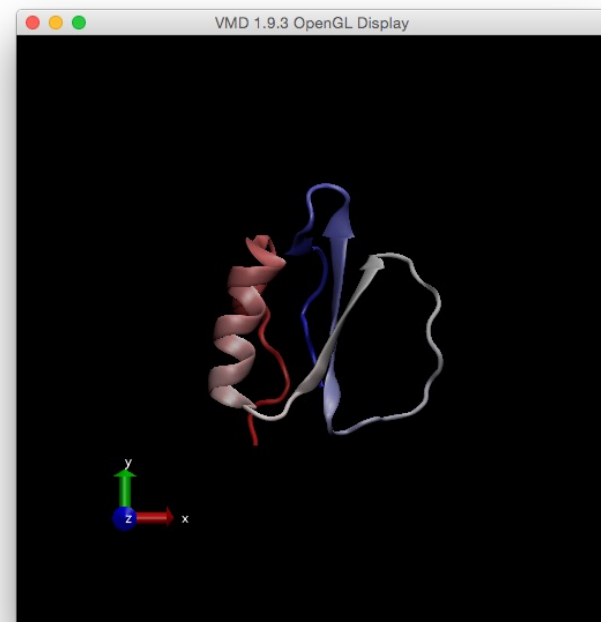
Select Coloring Method “Index”

Hit “Apply”

Before



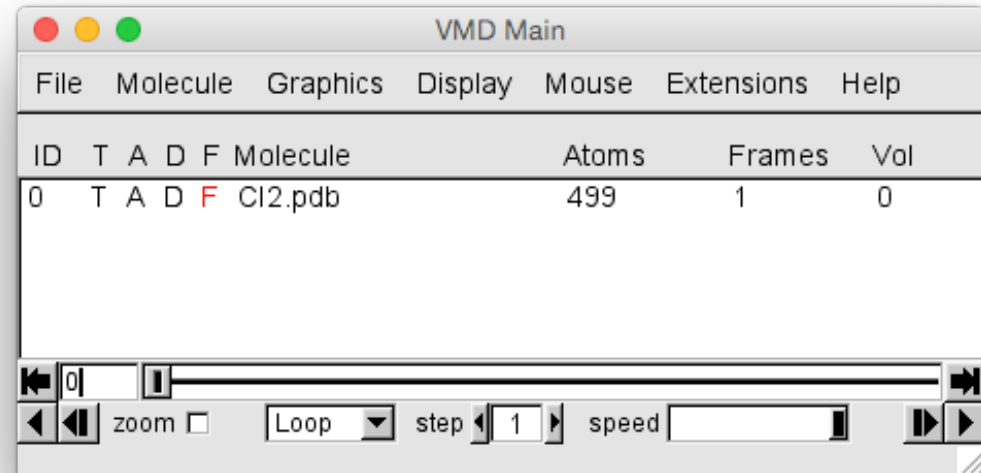
After



Load the Simulated Trajectory

Right click “Cl2.pdb” in the VMD Main window

Select “Load Data Into Molecule”



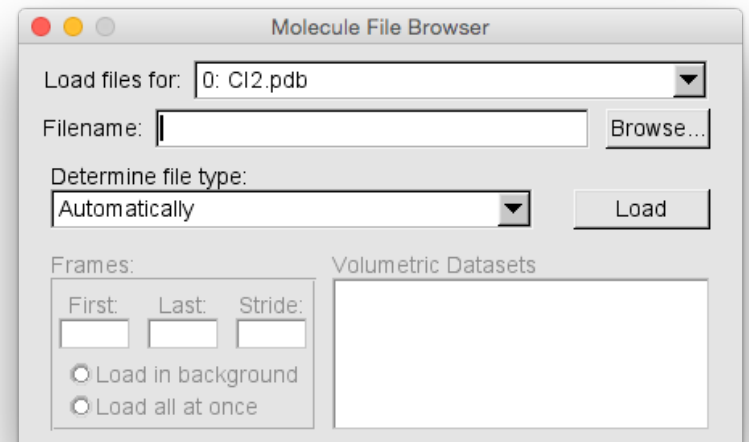
When Molecule File Browser opens,

Click “Browse”

Find the “vizfiles” folder

Select the file “fold.xtc”

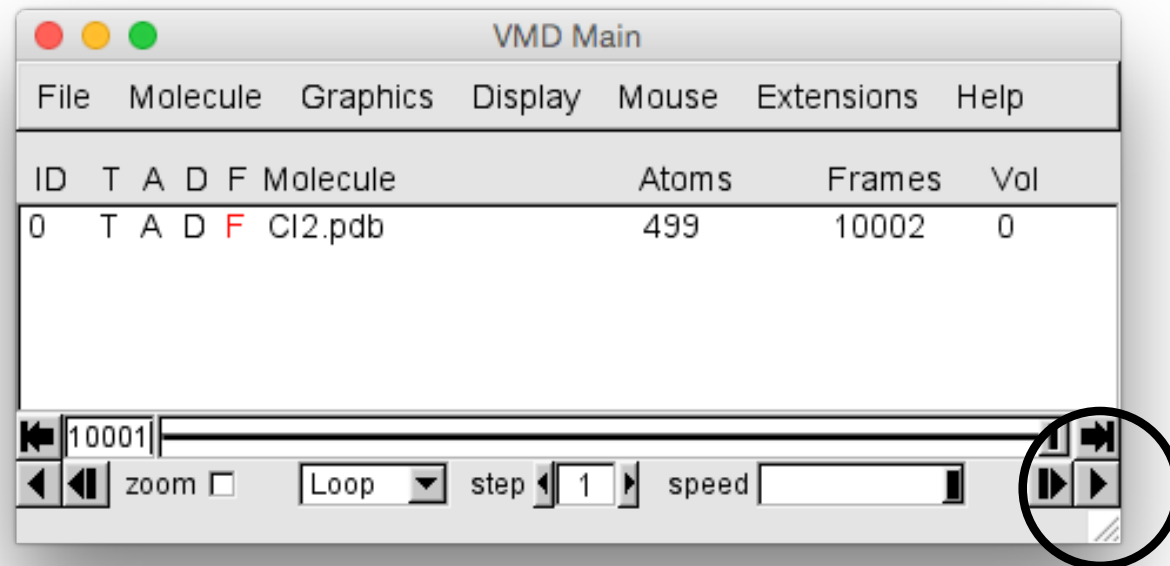
Click “Load”



While the trajectory is loading: Select Display window and hit “=” key to center the view

Replay the Movie

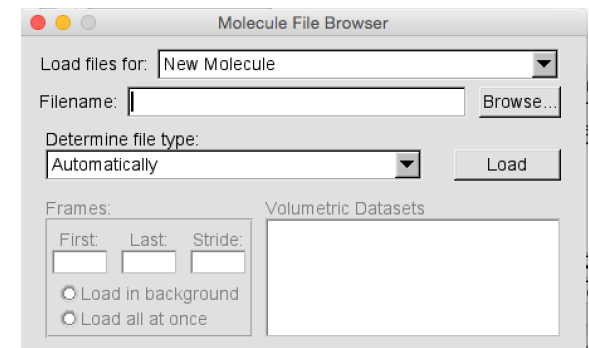
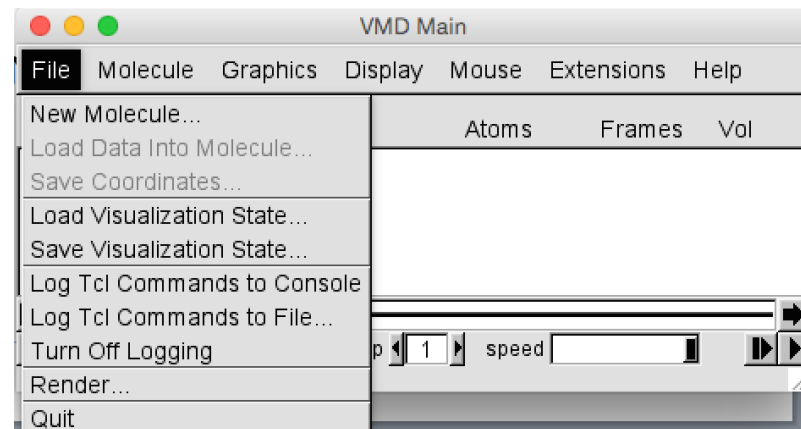
- Once the movie has stopped, you can restart it with the play button in the VMD Main window

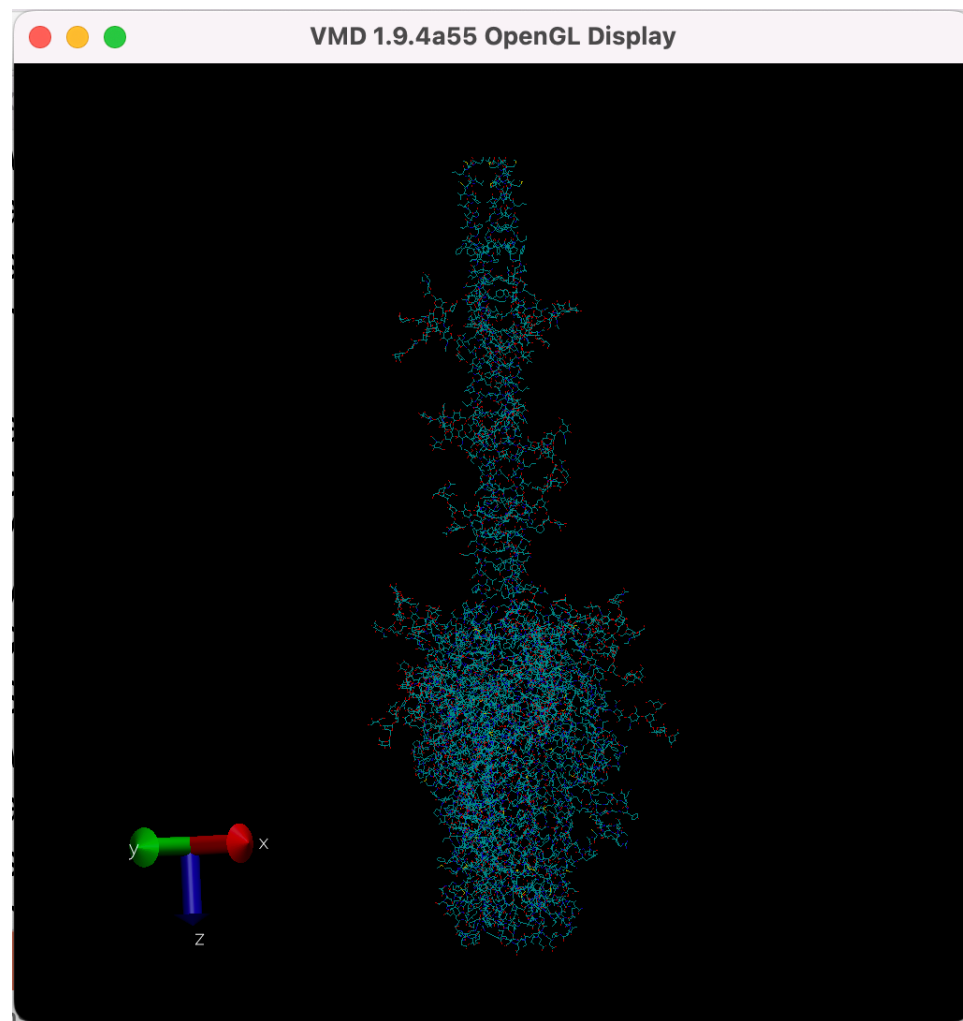
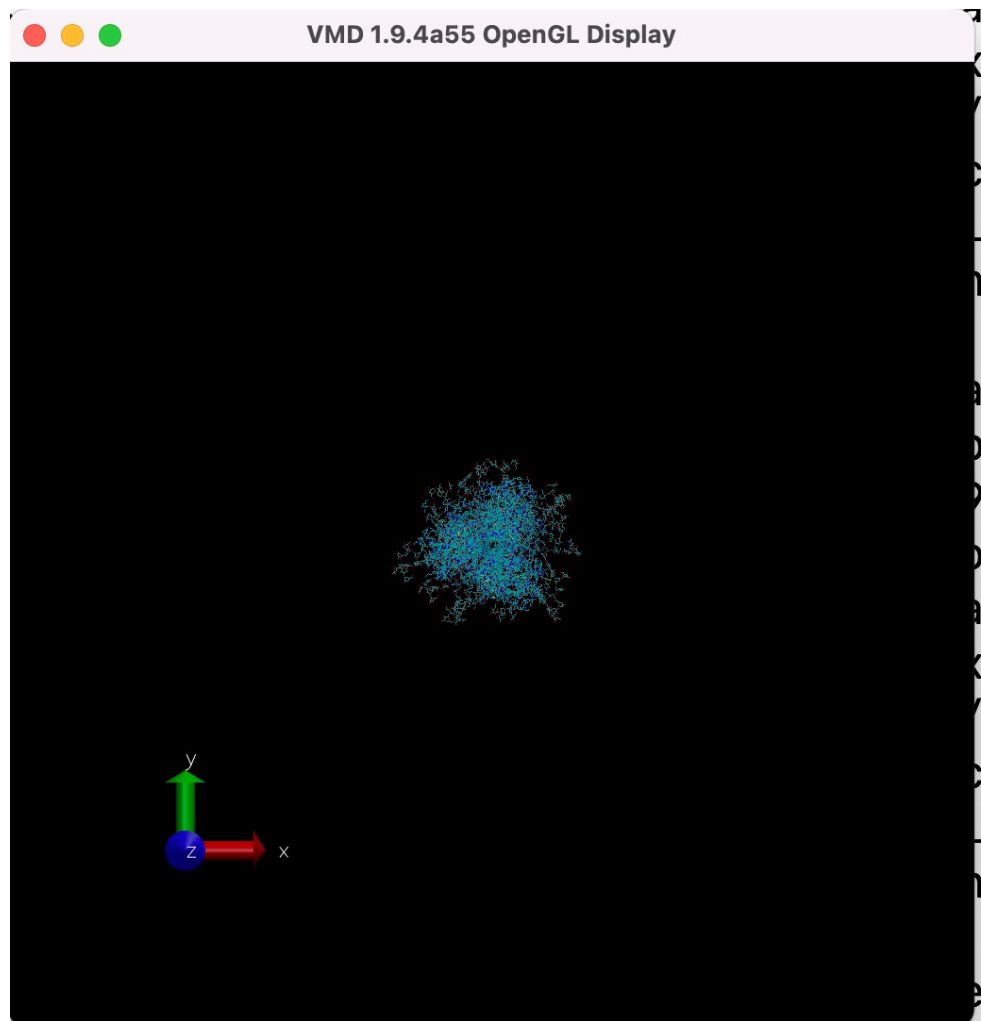


Exercise 3: The Spike Protein

Inspecting the Spike

- After opening VMD, load the protein structure.
- File->New Molecule
- When Molecule File Browser opens, select “Browse”. In the “vizfiles” folder, select the PDB file prefusion_with_glycans.pdb
- Select “Load”





We will work together

- Look at charged groups
- Select different groups and representations
- Consider orthographic and perspective views
- Calculate some distances and angles